

Definition 3.2: Dot Product

The dot product of two vectors $\vec{v} = \langle v_1, v_2 \rangle$ and $\vec{w} = \langle w_1, w_2 \rangle$ is defined as:

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2$$